

Trey Fitzpatrick

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Education

DUKE UNIVERSITY

Bachelor of Science in Engineering in Mechanical Engineering and Aerospace Engineering Certificate

Durham, NC

May 2026

GPA: 3.84/4.00 | Dean's List with Distinction Spring 2025, Fall 2024 | Dean's List Spring 2024

Coursework: Fluid Dynamics, Mechanical Design, Aircraft Performance, Heat and Mass Transfer, DSA

Work Experience

SPACEX

Mechanical Engineering Intern, Starship Launch Tower

Starbase, TX

May 2025-Aug 2025

- Responsible Engineer for installation, activation, and maintenance of tower hydraulic system, managing execution across mechanical, civil, electrical on \$500k+ project
- Owned/designed critical path component failure testing using pressurized system and tensile testing, presented findings to senior leadership leading to new component selection per recommendation
- Designed/built pneumatic system for 24/7 hydraulic pressure readiness, successfully keeping build on schedule
- Analyzed fluid networks to design, apply GD&T, and source hydraulic manifold at 40% reduced cost
- Designed and made manufacturing planning for T-0 fluid system upgrade, improving mass flow and analyzing for vibrational plume loading eliminating maintenance between flights

LARAMIE ENERGY

Mechanical and Process Engineering Intern

Denver, CO

Jun 2024-Aug 2024

- Developed app calculating piping losses to increase engineering efficiency by 90% and eliminate outsourcing
- Investigated cost-effective decarbonization solutions and determined most economical solution
- Performed boots-on-ground engineering, ensuring field implementation in-line with design and regulations

DUKE FIRST YEAR DESIGN PROGRAM

Engineering 101 Teaching Assistant

Durham, NC

Aug 2024-Present

- Mentored 5 interdisciplinary project groups, offering technical and interpersonal support for students throughout semester-long design project
- Reviewed and offered revisions on technical memos and design reviews

DUKE DEPARTMENT OF MECHANICAL ENGINEERING

Control Systems Teaching Assistant

Durham, NC

Aug 2024-Dec 2025

- Supported students with applied control systems, mentoring MagLev and record player controller development

Project Experience

DUKE AERO

Structures Sub-Team, High Power Rocketry Club

Durham, NC

Dec 2023-Present

- Designed and built a high-powered rocket to compete in Spaceport America Cup, winning the Jim Furfaro Award for Technical Excellence for best design and build out of 152 teams
- Performed FEA and manufactured bulkheads and airbrake housing on CNC mills and lathe
- Built airframe using composite processes for fiberglass and carbon fiber
- Designed and built composite camera shroud, recovered camera after rocket failure shortly after takeoff

DUKE AEROELASTICITY GROUP

Truss-Braced Wing, Independent Research Study

Durham, NC

Aug 2025-Present

- Leading a team to build a structural model of a novel wing concept for vibrational testing and analysis

Skills & Interests

Technical: Python, NX, HTML/CSS/JS, Fusion360, Ansys, Java, Sheet Metal, GD&T, Microsoft Office, Spanish

Manufacturing: Manual/CNC Mill, Manual Lathe, 3D Printing, Tube Bending, Weldments, Composites

Interests: Football, Golf, Backpacking, Fly fishing, Guitar